

Problem Description:

You are given an array A of Q integers and Q queries. In each query, you are given an integer i ($1 \leq i \leq N$).

Your task is to find the minimum index greater than i ($1 \leq i \leq N$) such that:

1. Sum of digits of A_i is greater than the sum of digits of A_j
2. $A_i < A_j$

If there is no answer, then print -1.

Constraints

$$1 \leq N, Q \leq 10^5$$

$$1 \leq A_i \leq 10^9$$

$$1 \leq Q_i \leq N$$

Input format

- The first line contains two numbers N and Q .
- The next line contains N numbers.
- Next Q lines contain Q queries.

Output format

Print the answer as described in the problem

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5 5
62 70 28 62 92
1
5
3
4
2

EXPECTED OUTPUT

2 -1 4 -1 -1

Test Case 2

INPUT (STDIN)

6 6
12 71 21 16 91
11
15
31
14
21

EXPECTED OUTPUT

-1 -1 -1 -1 -1 -1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(arr[x]<arr[y])

Test Case 2

KEYWORD

if(arr2[x]>arr2[y])

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

10

Test Case 2

TOKEN COUNT

310

Test Case 3

NLOC

57

Stack3.c +

```
1 #include <stdio.h>
2
3 int sumOfDigits(int n) {
4     int sum = 0;
5     while (n > 0) {
6         sum += n % 10;
7         n /= 10;
8     }
9     return sum;
10}
11
12int main() {
13    int N, Q;
14    if (scanf("%d %d", &N, &Q) != 2) return 0;
15
16    int arr[N + 1];
17    int arr2[N + 1];
18    for (int i = 1; i <= N; i++) {
19        scanf("%d", &arr[i]);
20        arr2[i] = sumOfDigits(arr[i]);
```

main | CH.SC.U4CSE25225 | Stack3.c

Ln 40, Col 1 Spaces: 4 UTF-8 CRLF C

```
22
23    for (int q = 0; q < Q; q++) {
24        int x;
25        scanf("%d", &x);
```

```
26     int ans = -1;
27     for (int y = x + 1; y <= N; y++) {
28         if (arr[x] < arr[y]) {
29             if (arr2[x] < arr2[y]) {
30                 ans = y;
31                 break;
32             }
33         }
```

```
34     }
35     printf("%d ", ans);
36 }
37 printf("\n");
38 return 0;
39}
40
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5 5

62 70 28 62 92

1

5

3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack3.exe

5 -1 5 5 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

6 6

12 71 21 16 91

11

15

31

14

21

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack3.exe

-1 -1 -1 -1 -1 -1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>